

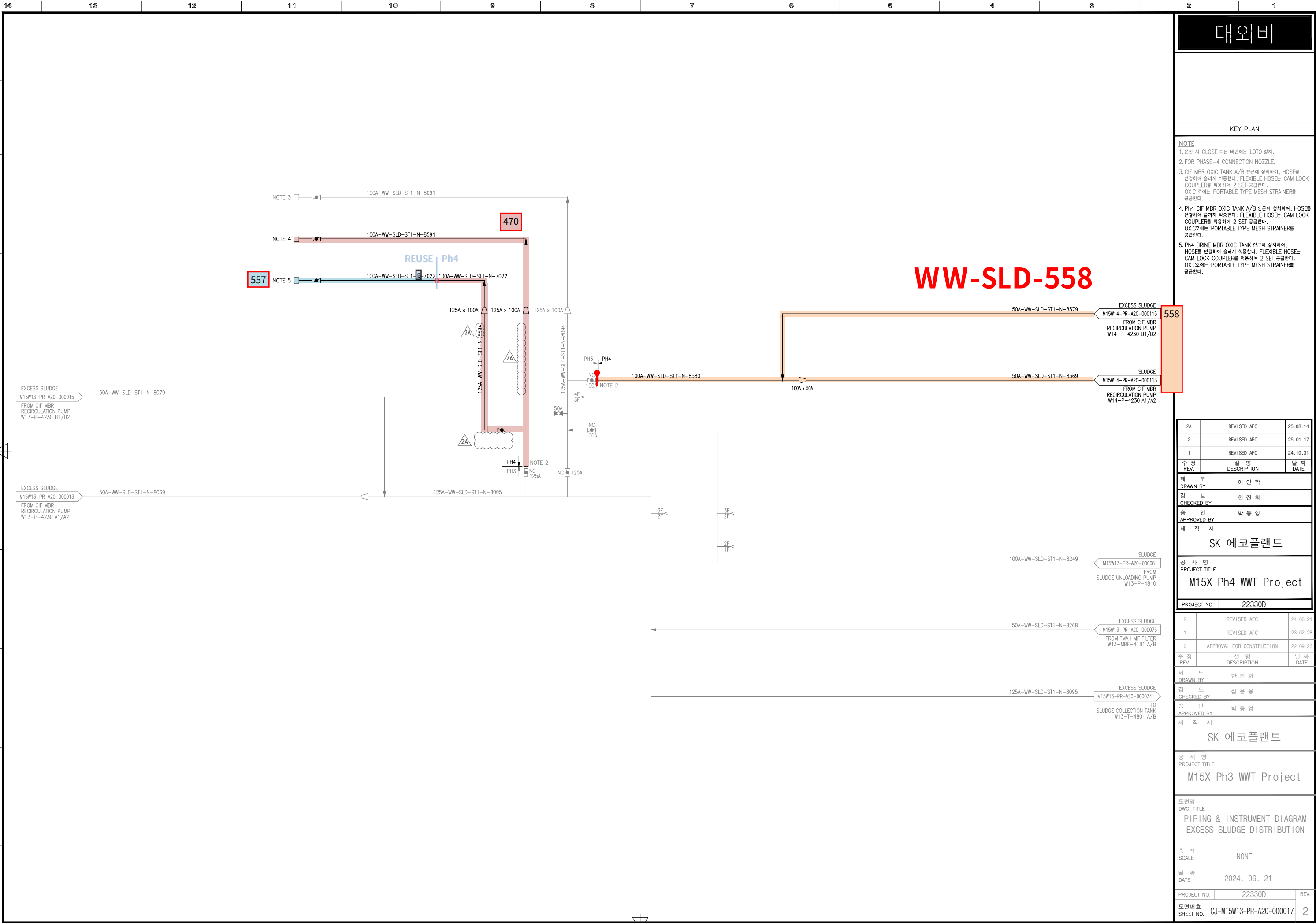
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

WW-SLD-558

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	558	3FA	50A-WW-SLD-ST1-N-8569-01	1 OF 3	0	CJ-M15W13-PR-A20-000017	2	
2	558	3FA	50A-WW-SLD-ST1-N-8569-02	2 OF 3	0	CJ-M15W13-PR-A20-000017	2	
3	558	3FA	50A-WW-SLD-ST1-N-8569-03	3 OF 3	0	CJ-M15W13-PR-A20-000017	2	
4	558	3FA	50A-WW-SLD-ST1-N-8579-01	1 OF 3	1	CJ-M15W13-PR-A20-000017	2	
5	558	3FA	50A-WW-SLD-ST1-N-8579-02	2 OF 3	0	CJ-M15W13-PR-A20-000017	2	
6	558	3FA	50A-WW-SLD-ST1-N-8579-03	3 OF 3	0	CJ-M15W13-PR-A20-000017	2	
7	558	3FA	100A-WW-SLD-ST1-N-8580-01	1 OF 1	0	CJ-M15W13-PR-A20-000017	2	
8	558	4FA	100A-WW-SLD-ST1-N-8580-01	1 OF 1	0	CJ-M15W13-PR-A20-000017	2	



대외비

KEY PLAN

NOTE

1. 운전 시 CLOSE 되는 배관에는 LOTO 설치.

2. FOR PHASE-4 CONNECTION NOZZLE.

3. CIF MBR OXIC TANK A/B 인근에 설치하여, HOSE를 연결하여 슬러지 착용한다. FLEXIBLE HOSE는 CAM LOCK COUPLER를 착용하여 2 SET 공급한다. OXIC 조에는 PORTABLE TYPE MESH STRAINER를 공급한다.

4. Ph4 CIF MBR OXIC TANK A/B 인근에 설치하여, HOSE를 연결하여 슬러지 착용한다. FLEXIBLE HOSE는 CAM LOCK COUPLER를 착용하여 2 SET 공급한다. OXIC 조에는 PORTABLE TYPE MESH STRAINER를 공급한다.

5. Ph4 BRINE MBR OXIC TANK 인근에 설치하여, HOSE를 연결하여 슬러지 착용한다. FLEXIBLE HOSE는 CAM LOCK COUPLER를 착용하여 2 SET 공급한다. OXIC 조에는 PORTABLE TYPE MESH STRAINER를 공급한다.

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도 이 인 학 DRAWN BY		
검 토 한 진 희 CHECKED BY		
승 인 박 동 영 APPROVED BY		
제 작 사 SK 에코플랜트		
공 사 명 PROJECT TITLE M15X Ph4 WWT Project		
PROJECT NO.		22330D
2	REVISED AFC	24.06.21
1	REVISED AFC	23.02.28
0	APPROVAL FOR CONSTRUCTION	22.09.23
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도 한 진 희 DRAWN BY		
검 토 심 문 용 CHECKED BY		
승 인 박 동 영 APPROVED BY		
제 작 사 SK 에코플랜트		
공 사 명 PROJECT TITLE M15X Ph3 WWT Project		
도면명 DWG. TITLE PIPING & INSTRUMENT DIAGRAM EXCESS SLUDGE DISTRIBUTION		
축척 SCALE		NONE
날짜 DATE		2024. 06. 21
PROJECT NO.		22330D
도면번호 SHEET NO.		CJ-M15W13-PR-A20-000017
REV.		2

ITEM NO.	W14-AD-4229 B
SERVICE	CIF MBR 2'nd OXIC TANK AIR DIFFUSER
Q'TY	110
TYPE	MEMBRANE (DISC)
CAPA.	L/min.ea
MAT'L	PP + EPDM

ITEM NO.	W14-T-4229 B
SERVICE	CIF MBR 2'nd OXIC TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	min
MAT'L	R.C + FRP

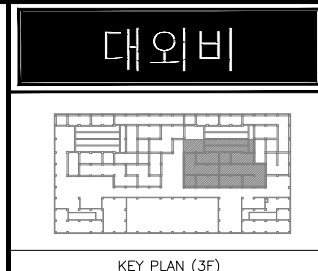
ITEM NO.	W14-T-4230 B
SERVICE	CIF MBR TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	hr
MAT'L	P.C + FRP

ITEM NO.	W14-T-4232 B
SERVICE	CIF MBR CLEANING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	min
MAT'L	C.S + FRP

ITEM NO.	W14-AD-4230 B
SERVICE	CIF MBR AIR DIFFUSER
Q'TY	ea
TYPE	MEMBRANE (DISC)
CAPA.	L/min.ea
MAT'L	PP + EPDM

ITEM NO.	W14-EJ-4231 B
SERVICE	CIF MBR EJECTOR UNIT
Q'TY	ea
TYPE	BY VENDOR
CAPA.	N/s.ea
MAT'L	BY VENDOR

ITEM NO.	W14-MBR-4230 B
SERVICE	CIF MBR MEMBRANE
CAPA.	m3/d
	15,000



- NOTE
1. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치함
 2. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성함
 3. MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP
 4. 온전 시 CLOSE 되는 배관에는 LOTO 설치
 5. CENTRIFUGAL PUMP
MAT'L 표기 : CASING / IMPELLER
 6. 4부 FRP LINING TANK SLEEVE
SUS304 + FRP LINING 적용
 7. PACKAGE VENDOR 선정 후 4부 SYSTEM
(모든 EQUIPMENT SPEC., LINE SIZE, MBR Nos., INSTRUMENT CONTROL SCHEME 등) 확정 예정
 8. 안전포기
 9. 축전포기
 10. 용량계 설치
DESIGN RANGE : -1~1 BAR

- INTERLOCK NO. 1-4231 B
(W14-P-4231 B1/B2 AUTO STOP)
RENOTE STATUS에 W14-T-4229 B LEVEL (LI-W2321) LOW LOW STOP.
 - OPERATION TIME CHART
- | PROCESS | SERVICE | B/W READY | B/W | RESTART |
|-----------|---------|-----------|------|---------|
| TIME(sec) | 24hr | 2 | 30 | 30 |
| XV-2331A | OPEN | - | - | - |
| XV-2331B | - | - | OPEN | - |
| XV-2321 | - | OPEN | - | OPEN |
| B/W PUMP | - | - | RUN | - |
4. 나머지 ON-OFF VALVE들도 동일하게 적용한다.
 13. CIF MBR TANK CRANE를 PACKAGE VENDOR에게 공급한다.
 14. DELETED.
 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 확정 예정
 16. MANHOLE 2개소 설치하여야 한다.
 17. SUPPLIED BY MBR PACKAGE VENDOR.
 18. HIGH POINT에 SIPHON BREAKER 설치함.
(CHECK VALVE의 경우 DUAL PLATE TYPE으로 적용)
 19. 개개의 세정용 CASSETTE CONNECTION 용도.
MBR CLEANING TANK 인근에 설치함
 20. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치함 (TAG NO. 아래 참조)
 - 150A-AIR-ST1-N-9528 : PG-W2381
 21. CIF MBR CLEANING TANK SHALL BE INSTALLED COVER WITH DUCT. OVERFLOW LINE(80A) TO DRAIN HEADER.
 22. 하도개 설치를 위한 FUTURE CONNECTIONS.
 23. PORT SIZE 350A. (TEE + 단관 + CAP 마감 : 단관 끝에는 W14-P-4231 PUMP의 CASING 일단 보다 높게 제작)
 24. AIR RELEASE VALVE
 25. SET POINT : 6BAR ~> 3 ~1 BAR
 26. SIZE IS 3,500mm x 1,600mm

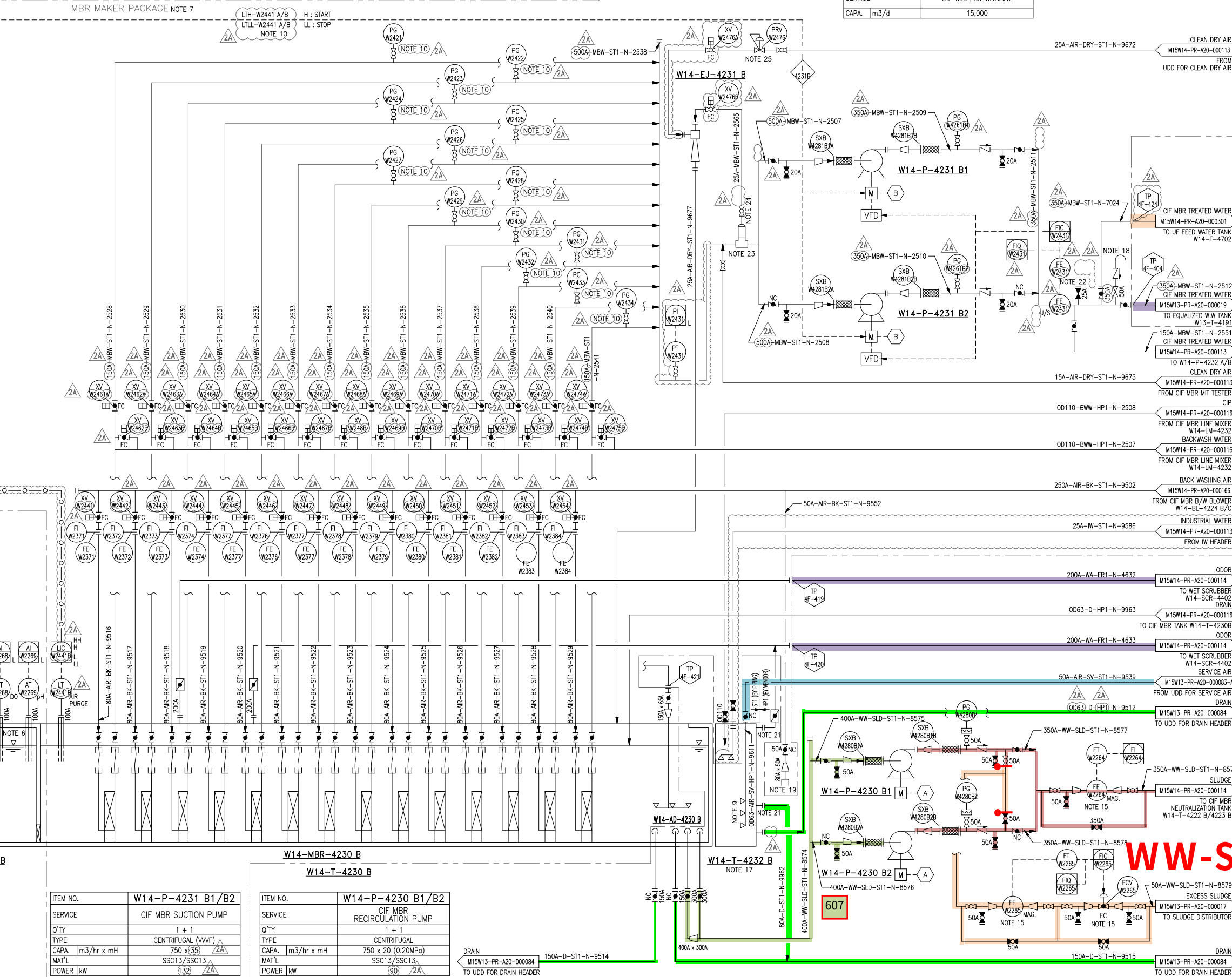
2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31

수정	REV.	실명	DESCRIPTION	날짜	DATE
553		이인학			
558		한진희			
558		박동영			

SK 에코플랜트	
사명	PROJECT TITLE
M15X Ph4 WWT Project	
도면명	WWT P&ID & INSTRUMENT DIAGRAM
CIF MBR DISCHARGE & CIRCULATION PUMP B (2/2)	
도면번호	SCALE
NONE	
날짜	2025. 08. 14
DATE	
PROJECT NO.	22330D
도면번호	REV.
SHEET NO.	CJ-M15W14-PR-A20-000115
2A	

ITEM NO.	W14-P-4231 B1/B2
SERVICE	CIF MBR SUCTION PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VWF)
CAPA.	m3/hr x mH
MAT'L	SSC13/SSC13
POWER	kw

ITEM NO.	W14-P-4230 B1/B2
SERVICE	CIF MBR RECIRCULATION PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL
CAPA.	m3/hr x mH
MAT'L	SSC13/SSC13
POWER	kw



WW-SLD 558

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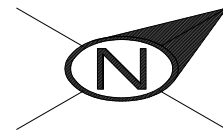
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CLIENT		 1	
CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WW-SLD		22330D	
LINE NO.	50A-WW-SLD-ST1-N-8569	SH'T NO. 01 / 02	REV. 1

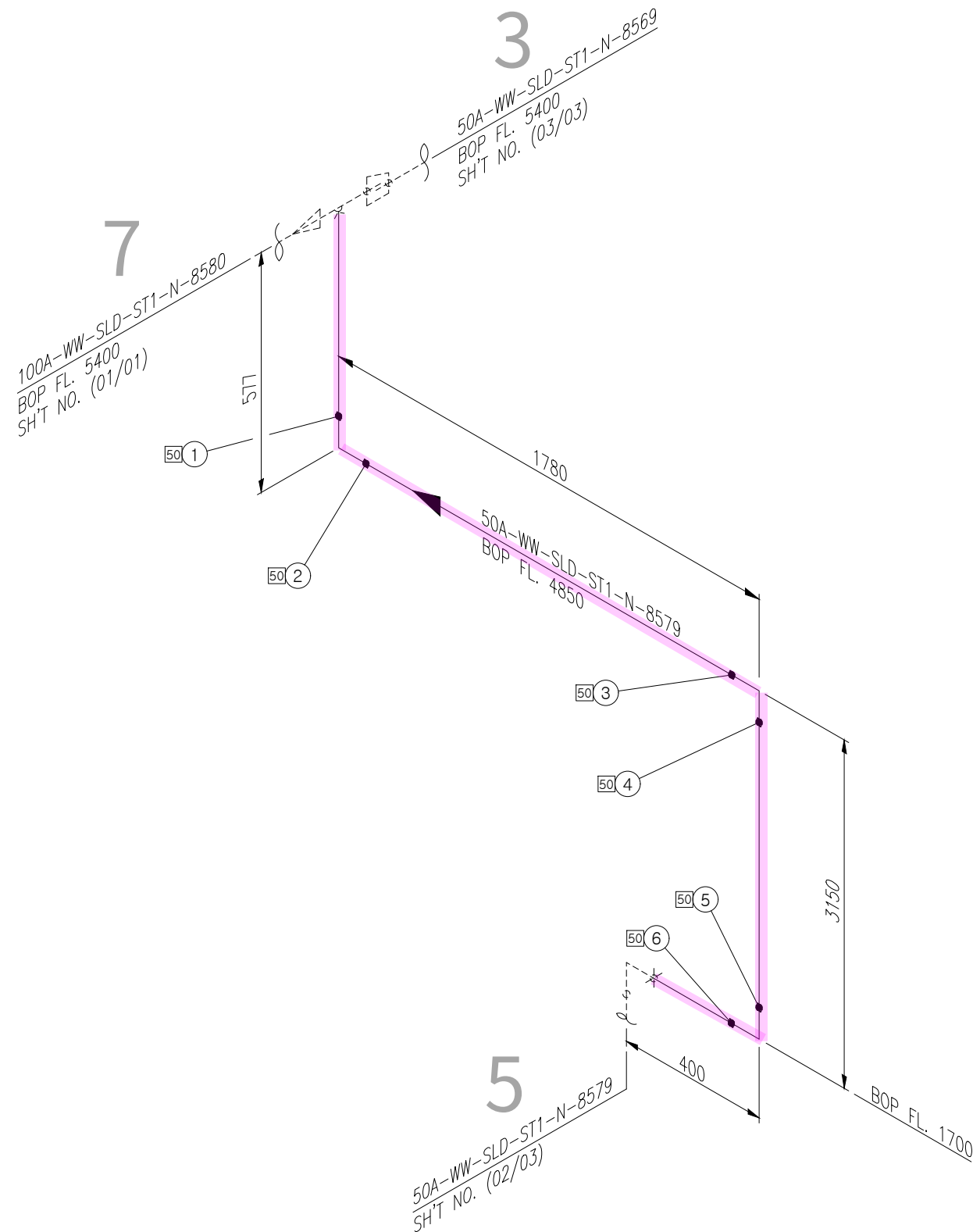
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1													
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	—	DESIGN	3.1	60	UNIT	3FA						
			OPERATING	2.0	AMB	REF. PLAN DWG.		1 25.03.31 REVISED AS MARKED 황래중 안남후 박동영					
						CJ-M15W14-PI-A30-030001-A			25.01.24 APPROVED FOR CONSTRUCTION 황래중 안남후 박동영				
						REF. P & ID CJ-M15W14-PR-A20-000113			DATE	DESCRIPTION	BY	CHK	APP'D



CLIENT				2	
CONTRACTOR				PROJECT NAME	
				M15X Ph4 Project	
COMMODITY		WW—SLD		JOB NO.	
				22330D	
영	LINE NO.	50A-WW-SLD-ST1-N-8569		SH'T NO.	REV.
D				02 / 03	0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1													
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7													
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH													
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1													

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	-	DESIGN	3.1	60	UNIT	3FA						
			OPERATING	2.0	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-030001-A							
						REF. P & ID							
					CJ-M15W14-PR-A20-000113			O 25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동	
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP		



<NOTE>
1. (*) FIELD ADJUST

[illegible]

CLIENT		6
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CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WW-SLD		22330D	
LINE NO.	50A-WW-SLD-ST1-N-8579	SH'T NO. 03 / 03	REV. 0

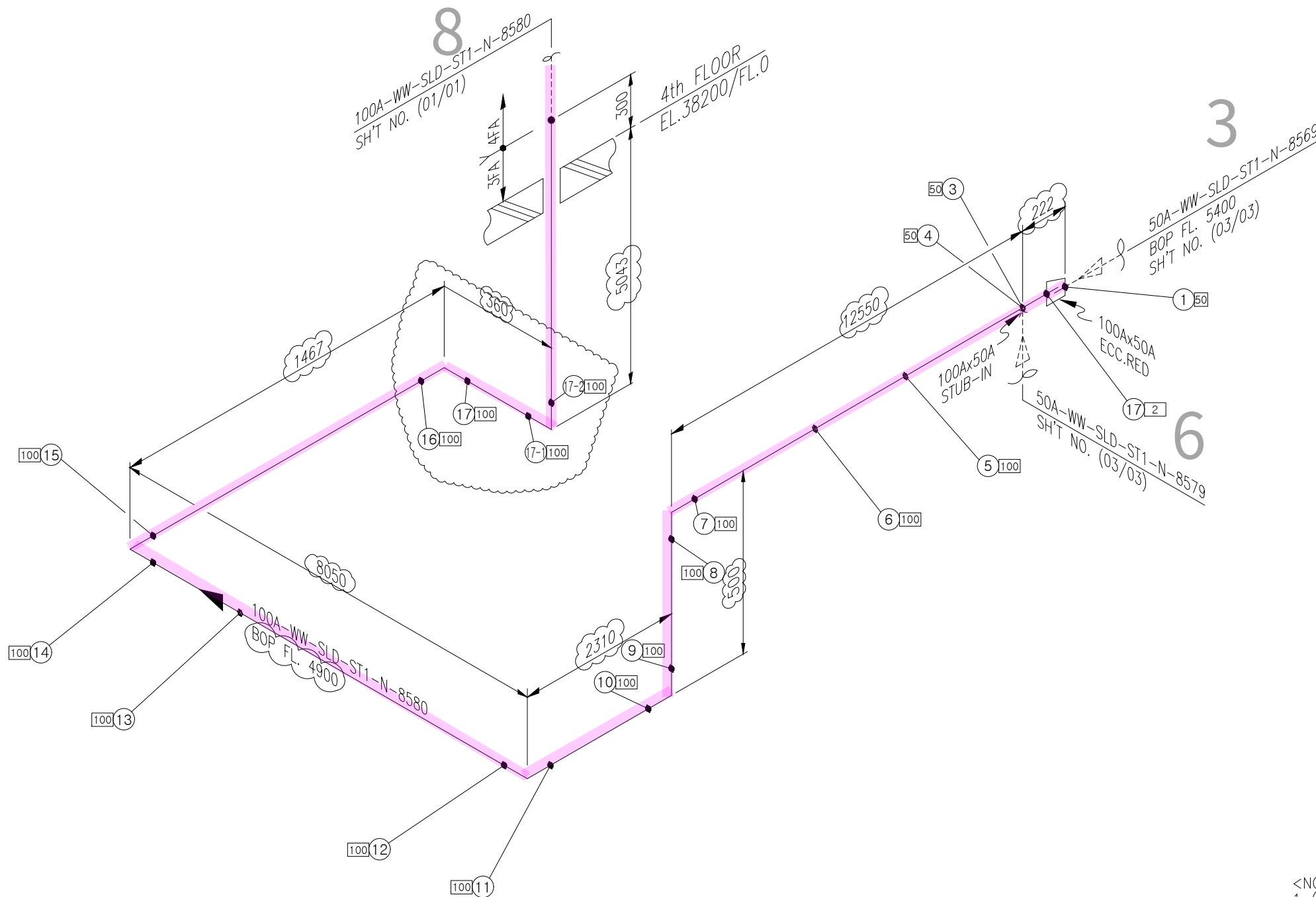
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	—	DESIGN	3.1	60	UNIT 3FA						
			OPERATING	2.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-030003-C		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000017						

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유포할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	100A	31.0M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	100A	6EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
RC	100Ax50A	1EA	REDUCER CON, STS304, ERW, BE - SCH 10S
STU	100Ax50A	1EA	STUB-IN BRANCH



<NOTE>
1. (*) FIELD ADJUST

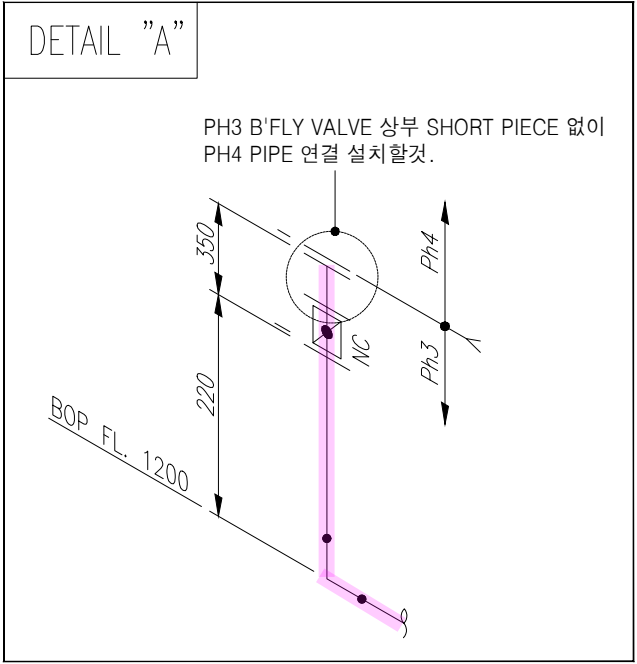
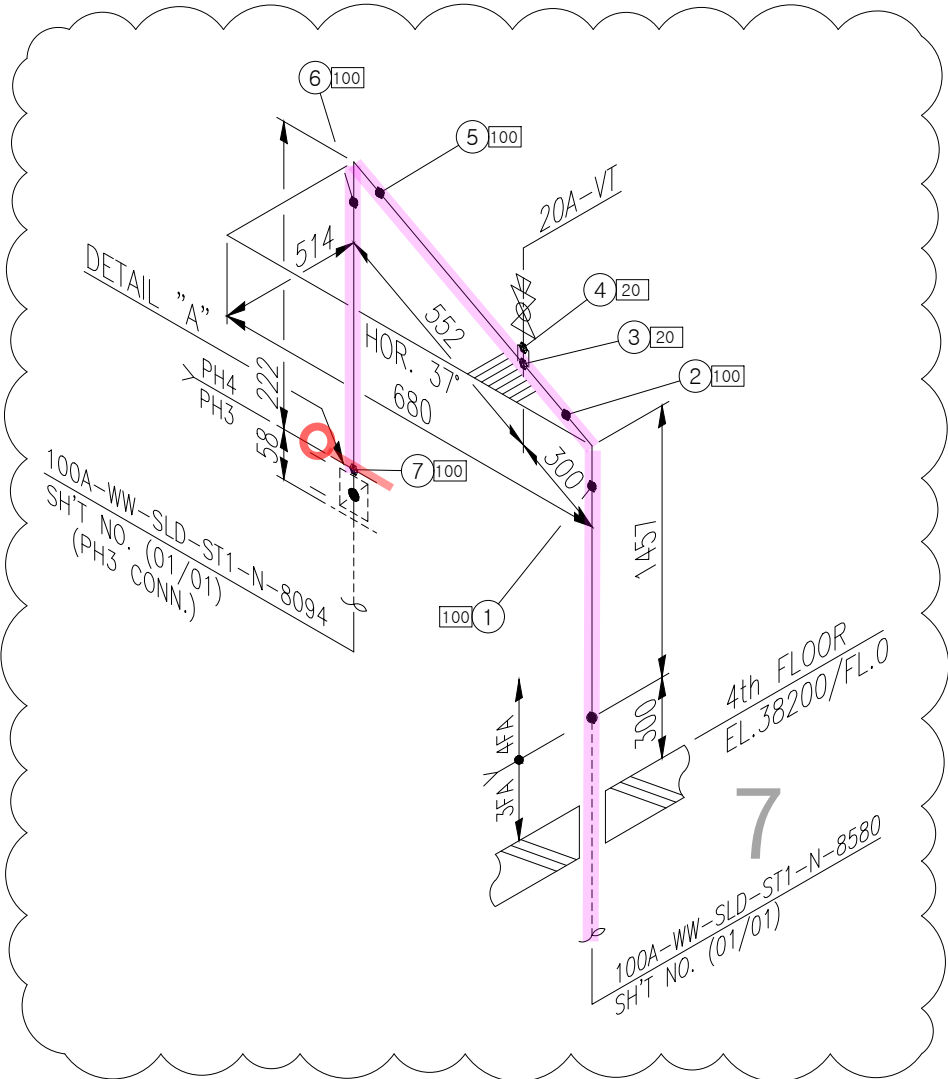
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/G1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR 	PROJECT NAME			
	THICK.	—	DESIGN	3.1	60	UNIT 3FA							M15X Ph4 Project			
			OPERATING	2.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-030003-C		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY WW—SLD		JOB NO.	
						22330D										
						REF. P & ID CJ-M15W14-PR-A20-000017	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	100A-WW-SLD-ST1-N-8580	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	100A	2.7M	PIPE, STS 304TP, ERW, BE – SCH 10S
E90	100A	2EA	ELBOW 90 DEG LR, STS304, ERW, BE – SCH 10S
HF	20A	1EA	COUPLING HALF, STSF304, SW – SCH 10S
NP1	20A	1EA	NIPPLE, STS 304TP, ERW, TOE – SCH 40S
PLG	20A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	1EA	BALL VALVE, KS 10K, SCR'D, BODY–STSF304 /
			TRIM–STS304 / SEAT–PTFE, FB, LEVER
G	100A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
BL	100A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)
F	100A	1EA	FLG SO, KS 10K, RF, STS304



<NOTE>
1. (*) FIELD ADJUST

CLIENT



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CONTRACTOR 		PROJECT NAME	
COMMODITY		JOB NO.	
WW-SLD		22330D	
LINE NO.	100A-WW-SLD-ST1-N-8580	SH'T NO.	REV.
		01 / 01	0

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	3.1	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	2.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-030003-C								COMMODITY	WW-SLD	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000017			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	100A-WW-SLD-ST1-N-8580	SH'T NO. 01 / 01	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				

장고 : 해당 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사나 유보자의 불법 사용으로 피해가 발생할 경우 유보자에게 민/형사상 책임을 요구할 수 있음